

Repeated Cross-Validation with Pooled-Seed Model Matrices

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Abstract

This report describes the revised cross-validation and fitting procedure for the behavioural models. For each subject, random block-based fold assignments are repeated multiple times. For every repetition and fold, a separate model is fitted, giving kR independent fits per model variant. Model stochasticity is handled by running the agent multiple times with different random seeds for the same parameter vector and pooling the simulated choices across seeds before estimating conditional psychometric and update matrices. The same train/test partitions are used for the Hybrid model and its special cases $\alpha = 0$ and $\alpha = 1$, ensuring that model comparisons are not confounded by different data splits.

Keywords

cross-validation, model fitting, pooled choices, update matrix, conditional matrix, behavioural modelling

1. Methods

1.1. Dataset and repeated fold assignments

Let

$$\mathcal{D} \tag{1}$$

denote the full dataset for one subject. The random assignment of data blocks into cross-validation folds is repeated

$$R = 10 \tag{2}$$

times.

For each repetition

$$r \in \{1, \dots, R\}, \tag{3}$$

the full dataset is randomly divided into

$$k \tag{4}$$

folds. In the present analysis, we use two-fold cross-validation:

$$k = 2. \tag{5}$$

For fold

$$f \in \{1, \dots, k\}, \tag{6}$$

the test set is denoted by

$$\mathcal{D}_{\text{test}}^{(r,f)}, \quad (7)$$

and the corresponding training set is

$$\mathcal{D}_{\text{train}}^{(r,f)} = \mathcal{D} \setminus \mathcal{D}_{\text{test}}^{(r,f)}. \quad (8)$$

Thus, r indexes the random fold assignment, while f indexes the held-out fold within that assignment.

1.2. Shared partitions across model variants

For each pair

$$(r, f), \quad (9)$$

the same training and test sets are used for all model variants. In particular, the Hybrid model and its two special cases

$$\alpha = 0 \quad \text{and} \quad \alpha = 1 \quad (10)$$

are evaluated using the same

$$\mathcal{D}_{\text{train}}^{(r,f)} \quad \text{and} \quad \mathcal{D}_{\text{test}}^{(r,f)}. \quad (11)$$

Here, $\alpha = 0$ corresponds to the BE special case, and $\alpha = 1$ corresponds to the SC special case. The partition information, including which sessions or blocks belong to each training and test set, is saved for every (r, f) . This ensures that differences in model performance are not caused by differences in the train/test split.

1.3. Empirical matrices

For each training set, we compute the empirical conditional matrix

$$C_{\text{train}}^{(r,f)}, \quad (12)$$

and the empirical update matrix

$$U_{\text{train}}^{(r,f)}. \quad (13)$$

Similarly, for each test set, we compute

$$C_{\text{test}}^{(r,f)}, \quad (14)$$

and

$$U_{\text{test}}^{(r,f)}. \quad (15)$$

The conditional matrix is computed from event counts. For current stimulus bin i and previous stimulus bin j , let

$$N_{ij} \quad (16)$$

be the number of valid post-correct trials where the current stimulus falls in bin i and the previous stimulus falls in bin j . Let

$$B_{ij} \quad (17)$$

be the number of those trials in which category B was chosen. The empirical conditional psychometric matrix is then

$$C_{ij} = \frac{B_{ij}}{N_{ij}}. \quad (18)$$

The update matrix is defined as the difference between the conditional post-correct psychometric matrix and the total post-correct psychometric curve:

$$U_{ij} = C_{ij} - \bar{\psi}_i, \quad (19)$$

where C_{ij} is the conditional probability of choosing category B at current stimulus bin i , conditioned on the previous stimulus bin j , and $\bar{\psi}_i$ is the total post-correct psychometric value at current stimulus bin i .

1.4. Pooled-seed model matrices

Let

$$\theta \quad (20)$$

denote a candidate model parameter vector. For a fixed training/test split (r, f) , the model is run

$$M = 10 \quad (21)$$

times with different random seeds. The parameter vector θ is kept fixed across these M runs; only the random seed changes.

For seed

$$m \in \{1, \dots, M\}, \quad (22)$$

let

$$\mathcal{S}_{\text{train}}^{(r,f,m)}(\theta) \quad (23)$$

denote the simulated choices generated by the model on the training set. The simulated choices are pooled across seeds:

$$\mathcal{S}_{\text{train}}^{(r,f)}(\theta) = \bigcup_{m=1}^M \mathcal{S}_{\text{train}}^{(r,f,m)}(\theta). \quad (24)$$

The model conditional and update matrices are then computed from this pooled simulated dataset. Importantly, the update and conditional matrices are not first computed separately for each seed and then averaged. Instead, the choices generated by the agent are pooled across all seeds, and a single conditional matrix and a single update matrix are estimated from the pooled choices.

Equivalently, for each seed m , define the model count matrices

$$N_{\text{train}}^{(r,f,m)}(\theta) \quad (25)$$

and

$$B_{\text{train}}^{(r,f,m)}(\theta), \quad (26)$$

where $N_{\text{train}}^{(r,f,m)}(\theta)$ counts the number of valid simulated trials in each current-previous stimulus bin pair, and $B_{\text{train}}^{(r,f,m)}(\theta)$ counts the number of category-B choices in those same bins.

Counts are summed across seeds:

$$N_{\text{model,train}}^{(r,f)}(\theta) = \sum_{m=1}^M N_{\text{train}}^{(r,f,m)}(\theta), \quad (27)$$

and

$$B_{\text{model,train}}^{(r,f)}(\theta) = \sum_{m=1}^M B_{\text{train}}^{(r,f,m)}(\theta). \quad (28)$$

The pooled-seed model conditional matrix is computed by elementwise division:

$$C_{\text{model,train}}^{(r,f)}(\theta) = B_{\text{model,train}}^{(r,f)}(\theta) \oslash N_{\text{model,train}}^{(r,f)}(\theta), \quad (29)$$

where $\oslash$ denotes elementwise division.

The corresponding pooled-seed model update matrix is

$$U_{\text{model,train},ij}^{(r,f)}(\theta) = C_{\text{model,train},ij}^{(r,f)}(\theta) - \bar{\psi}_{\text{model,train},i}^{(r,f)}(\theta), \quad (30)$$

where

$$\bar{\psi}_{\text{model,train},i}^{(r,f)}(\theta) \quad (31)$$

is the total post-correct psychometric value at current stimulus bin i , computed from the same pooled simulated choices.

The same pooling procedure is used for the test set after fitting.

1.5. Distance function

Let

$$A \quad (32)$$

be a model-predicted matrix and

$$B \quad (33)$$

be the corresponding empirical matrix. The distance between them is computed column by column.

For column j , define the set of valid row indices as

$$\mathcal{J}_j = \{i : A_{ij} \text{ and } B_{ij} \text{ are not NaN}\}. \quad (34)$$

A column is included only if it has at least one valid entry and the empirical column has non-zero squared norm. Therefore,

$$\mathcal{J} = \left\{ j : \mathcal{J}_j \neq \emptyset \quad \text{and} \quad \sum_{i \in \mathcal{J}_j} B_{ij}^2 > 0 \right\}. \quad (35)$$

The column-normalised matrix distance is then defined as

$$d(A, B) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \frac{\sum_{i \in \mathcal{I}_j} (A_{ij} - B_{ij})^2}{\sum_{i \in \mathcal{I}_j} B_{ij}^2}. \quad (36)$$

This distance gives each valid empirical column comparable weight, because the squared error in each column is normalised by the squared norm of the corresponding empirical column.

1.6. Training error for one repetition and fold

For each repetition r and fold f , the training error is computed by comparing the pooled-seed model matrix to the corresponding empirical training matrix.

If conditional matrices are used for fitting, the training error is

$$E_{\text{train}}^{(r,f)}(\theta) = d\left(C_{\text{model,train}}^{(r,f)}(\theta), C_{\text{train}}^{(r,f)}\right). \quad (37)$$

If update matrices are used for fitting, the training error is

$$E_{\text{train}}^{(r,f)}(\theta) = d\left(U_{\text{model,train}}^{(r,f)}(\theta), U_{\text{train}}^{(r,f)}\right). \quad (38)$$

1.7. Training objective

For each pair

$$(r, f), \quad (39)$$

a separate parameter vector is fitted. The objective is the fold-specific training error:

$$\theta_{r,f}^* = \arg \min_{\theta} E_{\text{train}}^{(r,f)}(\theta). \quad (40)$$

Thus, the optimisation is not performed on the average training error across folds. Instead, a separate optimisation is performed for every repetition and fold. Therefore, each model variant yields

$$kR \quad (41)$$

independent fitted parameter vectors per subject:

$$\{\theta_{r,f}^* : r = 1, \dots, R, \quad f = 1, \dots, k\}. \quad (42)$$

For conditional-matrix fitting, this objective can be written explicitly as

$$\theta_{r,f}^* = \arg \min_{\theta} d\left(C_{\text{model,train}}^{(r,f)}(\theta), C_{\text{train}}^{(r,f)}\right). \quad (43)$$

For update-matrix fitting, the corresponding objective is

$$\theta_{r,f}^* = \arg \min_{\theta} d\left(U_{\text{model,train}}^{(r,f)}(\theta), U_{\text{train}}^{(r,f)}\right). \quad (44)$$

The fitted training error for this repetition and fold is then

$$E_{\text{train}}^{(r,f)}(\theta_{r,f}^*). \quad (45)$$

1.8. Test error after fitting

After obtaining the fitted parameter vector

$$\theta_{r,f}^* \quad (46)$$

the model is evaluated on the corresponding test set

$$\mathcal{D}_{\text{test}}^{(r,f)}. \quad (47)$$

For each seed m , the fitted model generates simulated test choices

$$\mathcal{S}_{\text{test}}^{(r,f,m)}(\theta_{r,f}^*). \quad (48)$$

These choices are pooled across seeds:

$$\mathcal{S}_{\text{test}}^{(r,f)}(\theta_{r,f}^*) = \bigcup_{m=1}^M \mathcal{S}_{\text{test}}^{(r,f,m)}(\theta_{r,f}^*). \quad (49)$$

From the pooled simulated test choices, we compute

$$C_{\text{model,test}}^{(r,f)}(\theta_{r,f}^*) \quad (50)$$

and

$$U_{\text{model,test}}^{(r,f)}(\theta_{r,f}^*). \quad (51)$$

If conditional matrices are used, the test error is

$$E_{\text{test}}^{(r,f)}(\theta_{r,f}^*) = d\left(C_{\text{model,test}}^{(r,f)}(\theta_{r,f}^*), C_{\text{test}}^{(r,f)}\right). \quad (52)$$

If update matrices are used, the test error is

$$E_{\text{test}}^{(r,f)}(\theta_{r,f}^*) = d\left(U_{\text{model,test}}^{(r,f)}(\theta_{r,f}^*), U_{\text{test}}^{(r,f)}\right). \quad (53)$$

1.9. Final reported errors

The final mean training error is obtained by averaging over all repetitions and folds:

$$\bar{E}_{\text{train}} = \frac{1}{kR} \sum_{r=1}^R \sum_{f=1}^k E_{\text{train}}^{(r,f)}(\theta_{r,f}^*). \quad (54)$$

The final mean test error is

$$\bar{E}_{\text{test}} = \frac{1}{kR} \sum_{r=1}^R \sum_{f=1}^k E_{\text{test}}^{(r,f)}(\theta_{r,f}^*). \quad (55)$$

An alternative weighted estimate can also be computed. For the training error,

$$\bar{E}_{\text{train,weighted}} = \frac{\sum_{r=1}^R \sum_{f=1}^k n_{\text{train}}^{(r,f)} E_{\text{train}}^{(r,f)}(\theta_{r,f}^*)}{\sum_{r=1}^R \sum_{f=1}^k n_{\text{train}}^{(r,f)}}, \quad (56)$$

where $n_{\text{train}}^{(r,f)}$ is the number of training trials in fold (r, f) .

Similarly, for the test error,

$$\bar{E}_{\text{test,weighted}} = \frac{\sum_{r=1}^R \sum_{f=1}^k n_{\text{test}}^{(r,f)} E_{\text{test}}^{(r,f)}(\theta_{r,f}^*)}{\sum_{r=1}^R \sum_{f=1}^k n_{\text{test}}^{(r,f)}}. \quad (57)$$

In addition to the averaged errors, the fitted parameters and individual errors are saved for every pair (r, f) :

$$\theta_{r,f}^* \quad (58)$$

$$E_{\text{train}}^{(r,f)}(\theta_{r,f}^*), \quad (59)$$

and

$$E_{\text{test}}^{(r,f)}(\theta_{r,f}^*). \quad (60)$$

These saved values can be used for later statistical comparisons across model variants.

1.10. Variability across fits

The variability across the kR fits can be summarised using the standard deviation of the test errors:

$$\text{SD}_{\text{test}} = \sqrt{\frac{1}{kR-1} \sum_{r=1}^R \sum_{f=1}^k \left[E_{\text{test}}^{(r,f)}(\theta_{r,f}^*) - \bar{E}_{\text{test}} \right]^2}. \quad (61)$$

The standard error of the mean test error is

$$\text{SEM}_{\text{test}} = \frac{\text{SD}_{\text{test}}}{\sqrt{kR}}. \quad (62)$$

Analogous quantities can be computed for the training errors.

2. Interpretation

This procedure separates three sources of variability.

First, the same train/test partition is used for the Hybrid model and its special cases $\alpha = 0$ and $\alpha = 1$. Therefore, differences in model performance are not driven by differences in the underlying train/test split.

Second, model stochasticity is reduced by pooling simulated choices across

$$M = 10 \quad (63)$$

random seeds before estimating conditional psychometric and update matrices.

Third, variability due to random fold assignment is quantified by repeating the full cross-validation procedure

$$R = 10 \tag{64}$$

times.

Thus, each subject produces

$$kR \tag{65}$$

independent parameter estimates and corresponding train and test errors for each model variant. These individual fits can then be used for model comparison and statistical analysis.

3. Conclusion

The revised fitting procedure evaluates each parameter vector by pooling simulated choices across multiple model seeds and then computing a single conditional psychometric matrix and update matrix from the pooled choices. For each repetition and fold, a separate parameter vector is fitted, resulting in kR independent fits per subject and model variant. The same train/test partitions are used for the Hybrid model and its special cases $\alpha = 0$ and $\alpha = 1$, enabling direct comparison between model variants. Final training and test errors are computed by averaging the individual errors over all repetitions and folds.